SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0515** **Slot B**

COURSE OUTCOME COVERED
CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)***

Assessment Tool Weightage: 10%

QUESTIONS:5 **Total Marks 50** **Duration :60 Minutes**

Annexure A

Comparative Analysis

Code of Conduct:

I M. Pranay (192525382)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SFM)

Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

2. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

4. F1-Score Comparison

Two models:

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

5. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

ANSWERS

1. Depth Estimation Comparison — Stereo vs SFM

Formula:
Absolute Error = |Estimated Depth − Actual Depth|

Object	Actual	Stereo	Stereo Error	SFM	SFM Error
A	2.0	2.0	0.0 m	2.5	0.5 m
B	3.0	3.0	0.0 m	3.8	0.8 m
C	4.0	4.0	0.0 m	5.0	1.0 m

Evaluation:

- Stereo total absolute error = 0.0 m
- SFM total absolute error = 2.3 m
- Stereo mean absolute error = 0.0 m
- SFM mean absolute error = 0.767 m

Conclusion: Stereo Vision is more accurate because its estimated depths exactly match the actual depths.

2. Pattern Recognition Accuracy Comparison

Formula:
Accuracy = (Correct Predictions / Total Samples) × 100
Traditional System:
= (85 / 100) × 100
= 85%
Deep Learning System:
= (92 / 100) × 100
= 92%
Comparison:

- Traditional = 85%
- Deep Learning = 92%
- Improvement = 7 percentage points

Conclusion: Deep Learning performs better because it achieves higher recognition accuracy of 92%.

3. Precision–Recall Trade-off Analysis

Formulas:

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A

Precision = 70 / (70 + 30) = 0.70 = 70%

Recall = 70 / (70 + 20) = 0.778 = 77.8%

Model B

Precision = 85 / (85 + 25) = 0.773 = 77.3%

Recall = 85 / (85 + 25) = 0.773 = 77.3%

Model	Precision	Recall
A	70%	77.8%
B	77.3%	77.3%

Conclusion: Model B is more balanced because its precision and recall are equal at approximately 77.3%.

4. F1-Score Comparison

Formula:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Model A

F1 = 2 × (0.8 × 0.6) / (0.8 + 0.6)

= 0.96 / 1.4

= 0.686

Model B

F1 = 2 × (0.75 × 0.75) / (0.75 + 0.75)

= 1.125 / 1.5

= 0.75

Model	Precision	Recall	F1-Score
A	0.80	0.60	0.686
B	0.75	0.75	0.750

Conclusion: Model B is better because it has the higher F1-score (0.75) and provides a better balance between precision and recall.

5. Optical Flow Consistency
Formula:
Average Motion = Sum of motion values / Number of frames
Method A
Average = (4 + 5 + 6) / 3
= 15 / 3
= 5
Method B
Average = (6 + 8 + 9) / 3
= 23 / 3
= 7.67

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9
Average	5.00	7.67

Consistency:
- Method A changes: 4 → 5 → 6
- Method B changes: 6 → 8 → 9

Method A has smaller frame-to-frame changes and a more gradual variation.
Conclusion: Method A is more stable/consistent, while Method B has a higher average motion value.